

2016 New Year wishes to the American electorate. Essentially, your task is to organise the search engine query results on a timeline. What are the challenges in organising the Web search engine query results temporally?

- (6) You are asked to write a script that can list the set of processes currently running on your Linux machine and satisfy the criterion that they have been running for a time $> T$, where T is provided as an argument to the script, in seconds. This should include all the processes in your system, including those of other users and system processes.
- (7) You are given an array A of N integers. These integers represent the average yearly rainfall measured in centimetres, in your city for the last N years. Your function should output the frequency of the unique values of the rainfall measures contained in the input array. For instance, if you are given as input [5, 10, 5, 20, 10, 35, 5, 10], your function should output the following result:
 5 -> 3
 10 -> 3
 20 -> 1
 35 -> 1
 Essentially, for each unique value in the input array, output how many times it occurs in the given array. What is the time complexity of your algorithm?
 Now assume that the input array A cannot fit in main memory. In that case, how would you modify your approach to find the frequency distribution of array A ?
- (8) In bank cheques, a customer needs to write the amount of the cheque both in numbers and in words. For instance, if the cheque being issued is for Rs 3238 in numerical form, the customer also needs to provide a textual description of the amount – ‘Three thousand two hundred and thirty eight’. You are asked to write two functions: (a) function `cheque_itoa`, which takes as input a positive integer, and outputs a string which contains the textual description of the input integer; (b) function `cheque_atoi`, which takes as input a string containing the textual description of the cheque amount and returns the numerical integer of this amount.
- (9) You are given an array of N integers. You are asked to find the largest element which appears an even number of times in the array. What is the time complexity of your algorithm? Can you do this without sorting the entire array?
- (10) Given a stream of input integers in real-time, you are asked to find the minimum element in the stream at any point in time. What is the complexity of your algorithm? How would your solution change, (a) if you are asked to find both minimum and maximum elements in the stream at any point in time? (b) If you

are asked to find both the first minimum and second minimum in the stream at any point in time?

- (11) Given a singly linked list, you are asked to write a function to find the middle element of the linked list. You are not given the length of the list. How would you modify your solution if the given linked list is a singly linked circular list?
- (12) You are given four strings — S_1 , S_2 , S_3 and S_4 . S_1 is a large string and S_2 , S_3 , and S_4 are comparatively smaller. You are asked to find the smallest length sub-string ‘ t ’ of S_1 such that it contains S_2 , S_3 and S_4 as its sub-strings. For example, given string S_1 ‘BytheriversofBabylon’, and $S_2=er$, $S_3=of$ and $S_4=so$, your function should return ‘ersof’. How would you modify your solution if you are now told that the ordering of S_2 , S_3 and S_4 should be maintained in the sub-string ‘ t ’ (i.e., S_2 should appear before S_3 , and S_3 should appear before S_4 in sub-string ‘ t ’)?
- (13) You are given a large text ‘ T ’ and a small sub-string ‘ S ’. You are asked to find out whether sub-string S appears anywhere in T . What is the complexity of your solution? How would your solution change if you are asked to now find all the occurrences of ‘ S ’ in the large string ‘ T ’? Now, what is the time complexity of your algorithm?
- (14) You are given an HTML page containing texts, ads, images, banners, menu boxes, etc. Can you write a script to extract all the text contained in the Web page?
- (15) You are given a stream of incoming words. They are coming in at the interval of 1s. At any second, you are asked to find the 10 most repeated words seen in the stream so far. What is the complexity of your algorithm? How would you modify your solution, if you are also asked to print the occurrence frequencies for the 10 most repeated words seen in the stream up to that point in time? What is the time and space complexity of your solution?

If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Till we meet again next month, happy programming! 

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